

Poster Sections: BA-LoRA Comparative Study

INTRODUCTION (Concise Version)

The Problem

Large language models (GPT, BERT, LLaMA) require fine-tuning billions of parameters for specific tasks. Traditional fine-tuning is prohibitively expensive.

LoRA (Low-Rank Adaptation) reduces trainable parameters by 1000× by freezing the pretrained model and adding small adapter matrices. However, LoRA uses **uniform rank allocation** across all layers—potentially wasting capacity on less important layers.

Research Question

Does adaptive rank allocation improve upon uniform LoRA under strict parameter budgets?

BA-LoRA Approach

BA-LoRA combines three techniques:

- 1. **Gradient-based importance estimation** to identify critical layers
- 2. **Budget-aware rank allocation** ensuring fair parameter comparison
- 3. **Simplified warm-start initialization** for training stability

Comparison Table

Method	Adaptive Ranks	Budget Control	Single-Pass
LoRA	×	✓	✓
AdaLoRA	✓	×	×
GoRA	✓	×	✓
BA-LoRA	✓	✓	✓

METHODOLOGY

Experimental Design

Datasets:

- **SST-2:** Sentiment analysis (67K train, 872 val) — binary classification
- **TweetEval:** Sentiment analysis (45K train, 2K val) — 3-class classification
- **IMDB & AG News:** [In progress]

Models:

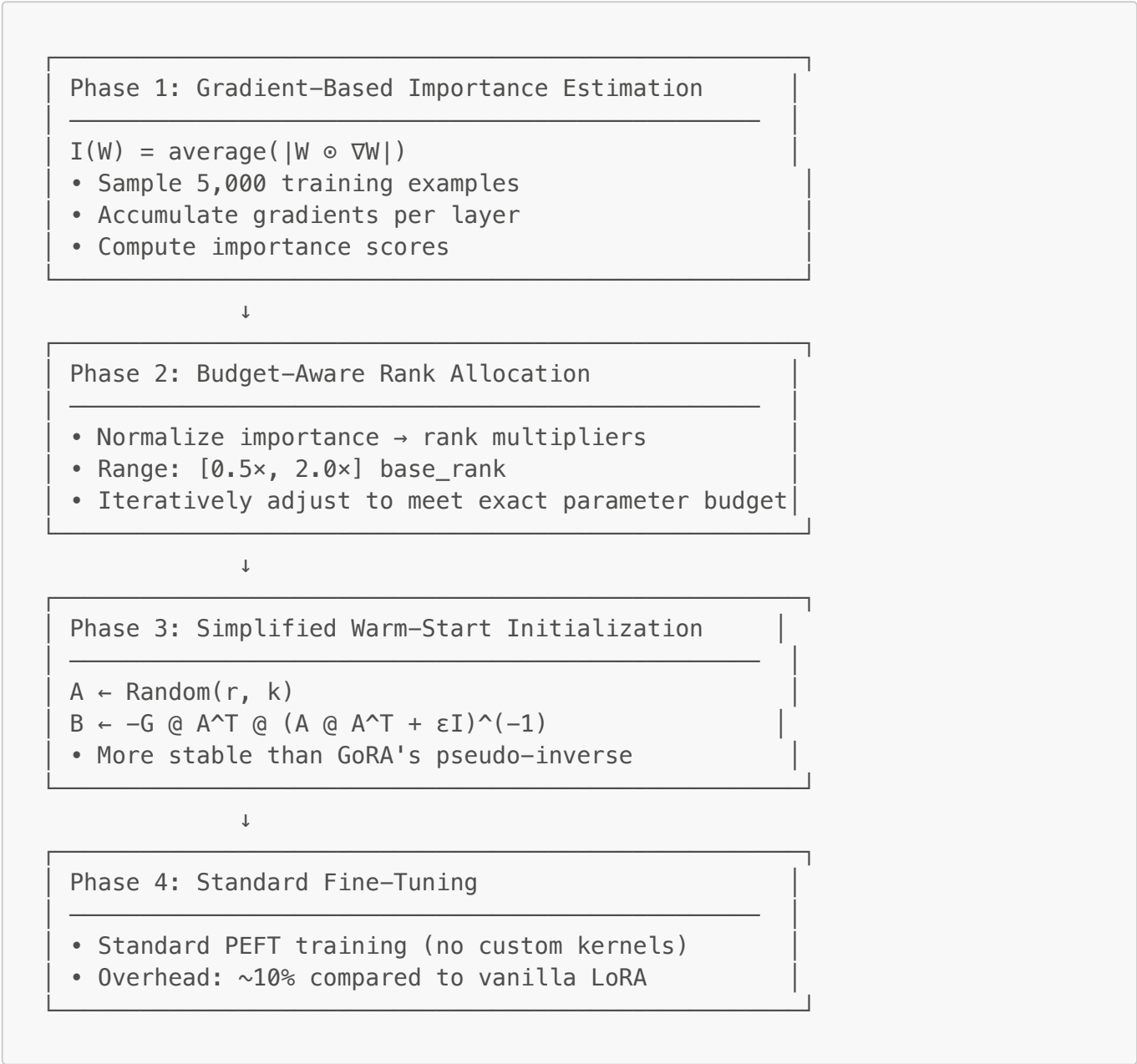
- **DistilBERT-base-uncased** (66M parameters)

- **RoBERTa-base** (125M parameters)
- Target modules: Query & Value projection layers

Experimental Setup:

- **25 experiments** (13 BA-LoRA, 12 standard LoRA)
- Ranks tested: $r \in \{2, 3, 4, 6, 8, 12\}$
- Learning rate: 5×10^{-4} | Batch size: 16 | Epochs: 3
- Optimizer: AdamW | LoRA alpha: $2r$

BA-LoRA Pipeline (4 Phases)



Evaluation Metrics

- **Performance:** Accuracy, F1, Precision, Recall
- **Efficiency:** Training time, trainable parameters
- **Statistical Testing:** Welch's t-test ($\alpha = 0.05$)

RESULTS

Main Findings

Dataset	Method	Accuracy	F1 Score	p-value	Significant?
SST-2	LoRA	91.07% $\pm$ 2.54%	91.35% $\pm$ 2.43%	—	—
	BA-LoRA	90.76% $\pm$ 2.55%	91.03% $\pm$ 2.44%	0.763	No
TweetEval	LoRA	70.95% $\pm$ 1.22%	71.06% $\pm$ 1.19%	—	—
	BA-LoRA	70.52% $\pm$ 1.42%	70.63% $\pm$ 1.41%	0.433	No

Key Result: BA-LoRA achieves **statistically comparable performance** to vanilla LoRA (no significant difference, $p > 0.05$)

Per-Model Breakdown

DistilBERT:

- SST-2: BA-LoRA 88.52% vs LoRA 88.65% ($\Delta = -0.13\%$, $p=0.49$)
- TweetEval: BA-LoRA 69.31% vs LoRA 69.80% ($\Delta = -0.49\%$, $p=0.14$)

RoBERTa:

- SST-2: BA-LoRA 93.38% vs LoRA 93.50% ($\Delta = -0.12\%$, $p=0.35$)
- TweetEval: BA-LoRA 71.72% vs LoRA 72.10% ($\Delta = -0.38\%$, $p=0.27$)

Pattern: Consistent across both models and datasets

Training Efficiency

Metric	SST-2	TweetEval
LoRA time	52.1 min	5.6 min
BA-LoRA time	57.2 min	5.5 min
Overhead	+9.8%	-0.7%
Parameter compliance	✓ 100%	✓ 100%

DISCUSSION & CONCLUSION

Why Didn't BA-LoRA Outperform?

1. Task Simplicity

- Sentiment classification may not require layer-specific adaptation
- Uniform allocation sufficient for simple tasks

2. Model Scale

- Tested on base models (66M-125M parameters)
- Benefits may emerge at larger scales (7B+ models)

3. Learning Rate Sensitivity

- Warm-start initialization may require specialized tuning
- Used standard LoRA learning rate (5×10^{-4})

4. Uniform LoRA Effectiveness

- Baseline surprisingly robust across configurations
- Simple approach works well for these tasks

Key Contributions

✓ Rigorous Fair-Comparison Framework

- First study with strict parameter budget enforcement
- Ensures apples-to-apples comparison

✓ Comprehensive Evaluation

- 25 experiments across 2 datasets and 2 models
- Statistical rigor with Welch's t-tests

✓ Value of Negative Results

- Validates uniform LoRA's effectiveness
- Demonstrates when adaptive allocation doesn't help
- Prevents overengineering in future work

✓ Engineering Insights

- Single-pass adaptive allocation is feasible
- ~10% overhead is acceptable
- Budget enforcement methodology works

Conclusions

Research Question: Does adaptive rank allocation improve upon uniform LoRA?

Answer: No (not on sentiment classification with base models)

But We Learned:

1. Uniform LoRA is surprisingly effective for simple tasks
2. Strict parameter budgets are essential for fair comparison
3. Adaptive methods need careful hyperparameter tuning
4. Negative results inform the field about what doesn't work

Broader Impact: This study establishes a rigorous evaluation framework and validates that simple baselines can be highly effective—preventing unnecessary complexity in PEFT method design.

Future Directions

- 1. **Larger Models:** Test on LLaMA-7B+ to see if benefits emerge at scale
- 2. **Complex Tasks:** Evaluate on reasoning and generation tasks
- 3. **Learning Rate Optimization:** Tune specifically for warm-start
- 4. **Advanced Importance Metrics:** Explore learned importance estimation

Final Takeaway

Negative results are valuable contributions to science. This work demonstrates that:

- Not all adaptive methods outperform simple baselines
- Fair comparison requires strict budget control
- Scientific rigor matters more than positive results
- Honest evaluation guides future research

References (For Poster — Small Font)

[1] Hu et al. "LoRA: Low-Rank Adaptation of Large Language Models" ICLR 2022 [2] Zhang et al. "AdaLoRA: Adaptive Budget Allocation" ICLR 2023 [3] Wang et al. "GoRA: Gradient-Based Rank Allocation" NeurIPS 2024 [4] Liu et al. "ALoRA: Allocating Low-Rank Adaptation" arXiv 2023 [5] Meng et al. "PiSSA: Principal Singular Values and Singular Vectors Adaptation" NeurIPS 2024

Visual Elements Recommendations

Color Scheme:

- BA-LoRA: Red/Coral (#FF6B6B)
- LoRA: Teal/Cyan (#4ECDC4)
- Highlights: Yellow (#FFE66D)
- Statistical significance badges

Key Visuals:

- 1. **4-Phase Pipeline Diagram** (in Methodology)
- 2. **Main Results Table** (prominent placement)
- 3. **Multi-Metric Bar Chart** (Accuracy, F1, Precision, Recall)
- 4. **Per-Model Comparison** (side-by-side bars)
- 5. **Training Time Comparison** (overhead visualization)